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Magnetostratigraphy of Paleogene sediments from northern Qaidam Basin, China: Implications for tectonic uplift and block rotation in northern Tibetan plateau

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Abstract

We have carried out a detailed magnetostratigraphic study of Tertiary sediments in the northern margin of the Qaidam Basin, northern Tibetan plateau. A total of 400 paleomagnetic samples (with an interval of about 2 m) were collected from the Shanggancaigou and Xiagancaigou Formations in the Hongsanhan section. Stepwise thermal demagnetization was used to isolate the magnetic components in these samples. A higher temperature component (HTC) was separated between 300 and 680 °C from most of samples. Positive fold and reversal tests suggest a primary origin for the HTC. A magnetic polarity sequence is established and then compared with the geomagnetic polarity time scale. Ostracoda assemblages and also a fission track age from an adjacent section suggest a Tertiary age of the Shanggancaigou and Xiagancaigou Formations in the Hongsanhan area. The magnetozones in the Xiagancaigou Formation may correspond to subchrons C16n-C18 ranging from >40 to 35.5 Ma (Mid-Eocene), and the Shanggancaikou Formation to subchrons C8.2n-C15 (age from 35.5 to 26.5 Ma Late Eocene to Late-Mid Oligocene). The accumulation of sandy conglomerate and conglomerate layers on top of the Shanggancaikou Formation, and an abrupt increase in sedimentation rates are probably resulted from the main uplift of the north Qaidam at about 30 Ma. When compared with coeval directions expected from Eurasia, the mean declinations indicate a slight vertical-axis rotation of the northern Qaidam Basin since the Mid-Eocene. Sinistral shear strain may be concentrated on the Altyn Tagh Fault (ATF).
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Keywords: Magnetostratigraphy; Qaidam Basin; Eocene; Oligocene; Rotation

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1. Introduction

Continental collision between India and Asia has resulted in crustal thickening and uplift of the Tibetan plateau. The timing and mechanisms of the Tibetan plateau growth are still poorly understood because of lack of widespread and precise age constraints on the crustal deformation and sedimentation in and around the plateau. In the northern Tibetan plateau, uplifting resulted from erosions led to huge accumulation of Cenozoic sediments that shed into foreland and hinder basins. Such process provided a useful method to study the relationships between mountain building, rock denudation, sediment deposition, and climatic changes. Age constraints on the Cenozoic terrestrial sediments, however, are usually hindered by limited fossil records or shortage of absolute dating in volcanic horizons.

Recently, in order to date Tertiary sediments and related uplift history, magnetostratigraphic studies have been carried out on the Tertiary basins adjacent to the Kunlun and Qilian Shan fold belts in the Tibetan plateau: the Linxia Basin [1,2], Guide Basin [3], Lanzhou Basin [4], Xinning-Minhe-Longzhong Basin [5], Yecheng Basin [6], Hoh Xil Basin [7] and Tianshan Mountain [8,9]. However, most of these studies were mainly limited to the Pliocene and younger deposits, and to the sediments in the foreland basins of the northern Tibetan plateau. No studies on the Tertiary magnetostratigraphy have been carried out in the Qaidam Basin, where the volumes and masses of sediments were eroded from the surrounded mountain ranges and the hinterland of the Tibetan plateau since the Tertiary. For better understanding of the interaction of tectonics and erosion and the early uplift history of the northern Tibetan plateau, we carried out a magnetostragraphic study on the Paleogene sediments from the northern Qaidam Basin, which provides the first detailed constraint on the age of the Tertiary sedimentary sequence, This result will be used further to discuss the uplift history

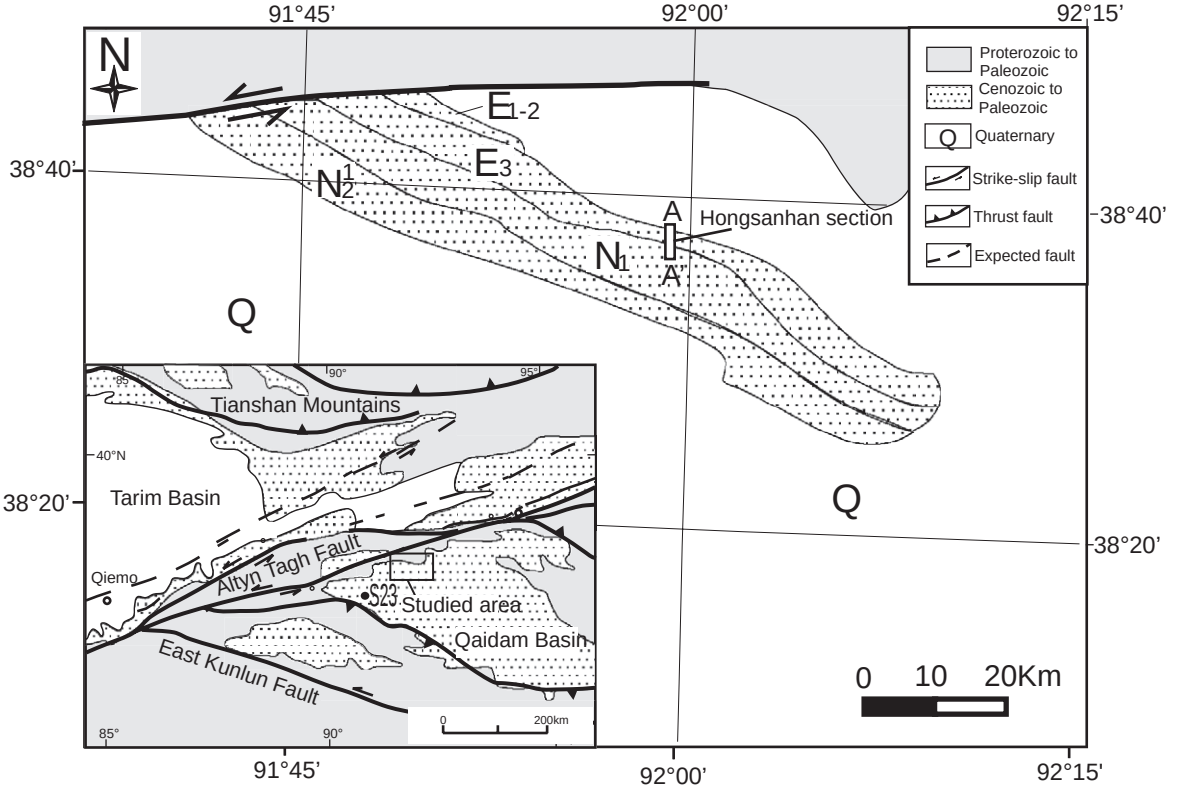


Fig. 1. Simplified geological map of the northwestern Qaidam Basin, showing the localities of paleomagnetic sampling sections. The locality of the drill core (No. S23) yielding FT data is marked by solid symbol.

of the northern Tibetan plateau and tectonic rotation of the Qaidam Basin.

2. Geological setting and paleomagnetic sampling

The Qaidam Basin is the largest Cenozoic sedimentary basin in the hinterland of the northern Tibetan plateau with an area of 1.2×10^5 km^2, it is limited by the Altyn Tagh fault to the north and northwest, and bounded by the Qilian and Kunlun fold belts to the northeast and to the south respectively. Thrusting tectonics and erosion-deposition are interdependent. Therefore, the Qaidam Basin is sensitive to the effects of uplift of the northern Tibetan plateau and the environmental evolution since the Cenozoic.

There is a consensus that terrestrial facies of deltaic and lacustrine type dominanted the Early Tertiary, and changed to lacustrine and fluvial in the Late Tertiary in the Qaidam Basin. However, the age of the Tertiary sediments in the Qaidam Basin is still in dispute, e.g. the age of the Xiagancaigou Formation is supposed to the Oligocene, and the Shanggancaigou Formation to the Early Mid Miocene by the regional stratigraphic correlation [10]. Liu et al. (1997) inferred that the lower part of the Xiagancaigou Formation may reach the Paleocene based on the fission track (FT) dating of gypsum (53.68 ± 5.24 Ma) from a drill core (No. S23) [11] located to the west of the studied section (Fig. 1). They also suggested that the Xiagancaigou Formation is Late Paleocene to Early Mid Eocene in age (59–43 Ma), and the Shanggancaigou

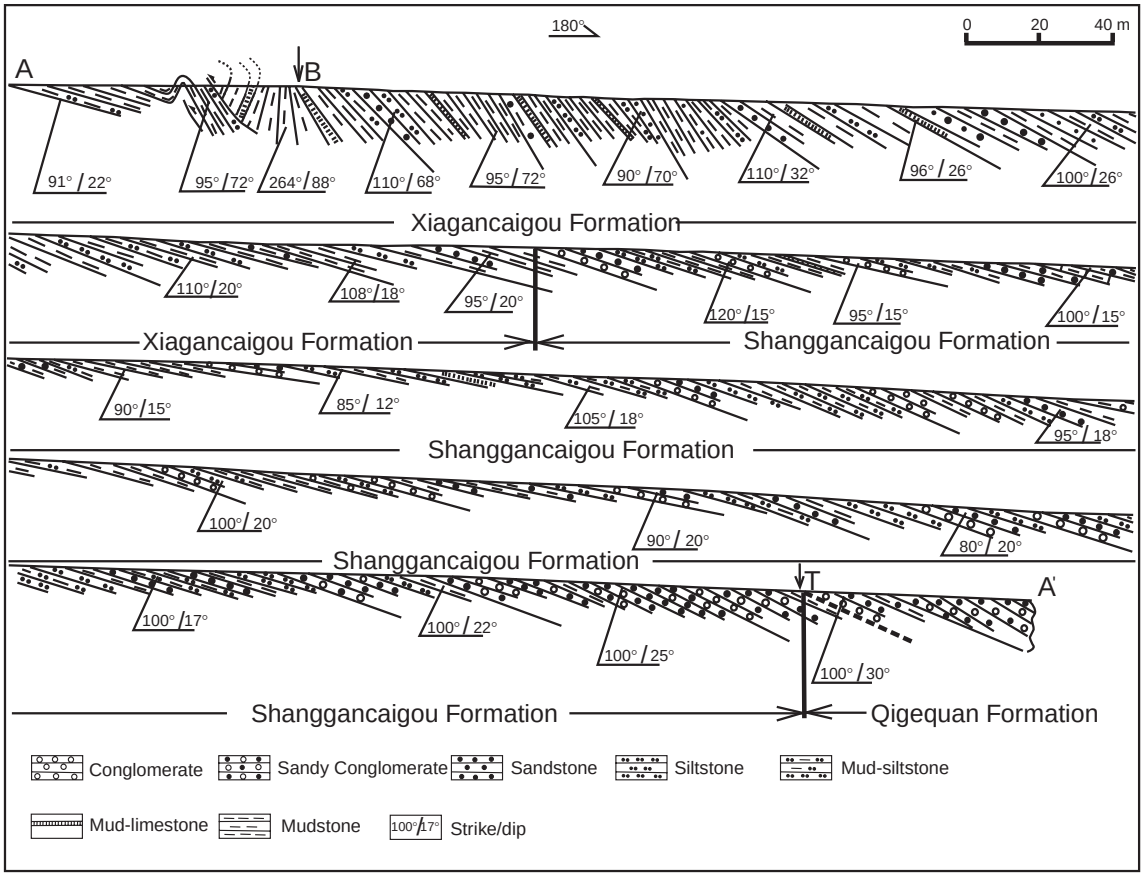


Fig. 2. Measured geological section through Paleogene at Hongsanhan. Dips are consistent between the upper part of Xiagancaigou and Shanggancaigou Formations, but become steeply in the lower part of Xiagancaigou Formation. Arrows show top (T) and bottom (B) of magnetostratigraphic section from this study.

Formation is Late Eocene to Oligocene in age based on the correlation of $\delta^{18}O$ with marine isotopic stages.

The sampling section is situated in the northern part of the Qaidam Basin, and to the south of the Altyn Tagh fault (Fig. 1). By lithologic correlation with the drill core (No. S23) (Fig. 1), the section lacks lower parts (about 500 m thick) of the Xiagancaigou Formation. It displays the most continuous exposure of suitable fine-grained lithologies with the bedding attitudes varying 24°–66° westward and southward dips (Fig. 2). The Paleogene sediments have a thickness of approximately 764 m in the Hongsanhan section, which consists of the Xiagancaigou and Shanggancaigou Formations.

The Xiagancaigou Formation with 368-m thick consists of gray mudstone, siltstone intercalated with sandstone and gypsum layers, which was mainly deposited in the salt lacustrine environments. No fossils were found in the studied section, however, the presence of Ostracoda assemblages such as *Austrocypris* sp., *Eucypris mutilis* F. Yang and Sun, *Cyprinotus giganto-triangulatus-Qaibeigouia reniformis* in the adjacent areas of the Qaidam Basin is characteristic of the Late Eocene to Oligocene for the Xiagancaigou Formation (Fig. 3) [12–14]. The overlying 396-m thick Shanggancaigou Formation can be divided into two parts. The lower part mainly consists of brown mudstone and siltstone, which was generally deposited in lacustrine environments. The upper part is mainly made up of brown sandstone and siltstone intercalated with more than 10 conglomerate layers, indicating a deviation in deposition environment from lacustrine facies. Three fossil layers including Ostracoda (②: *Hemicyprinotus, Hemicyprinotus Valvaetumidus Mandelstam*, ③: *Mediocypris, Mediocypris Candonaeformis*, ④: *Cyprinotus*) were found in the upper part of the Shanggancaigou Formation in the studied section (Figs. 3 and 7). These Ostracoda fossil assemblages imply that the Shanggancaigou Formation is an Oligocene to Middle Miocene age [12–14]. These fossil assemblages revealed in the Xiagancaigou and Shanggancaigou Formations provide a first order age constrains for this magnetostratigraphic study.

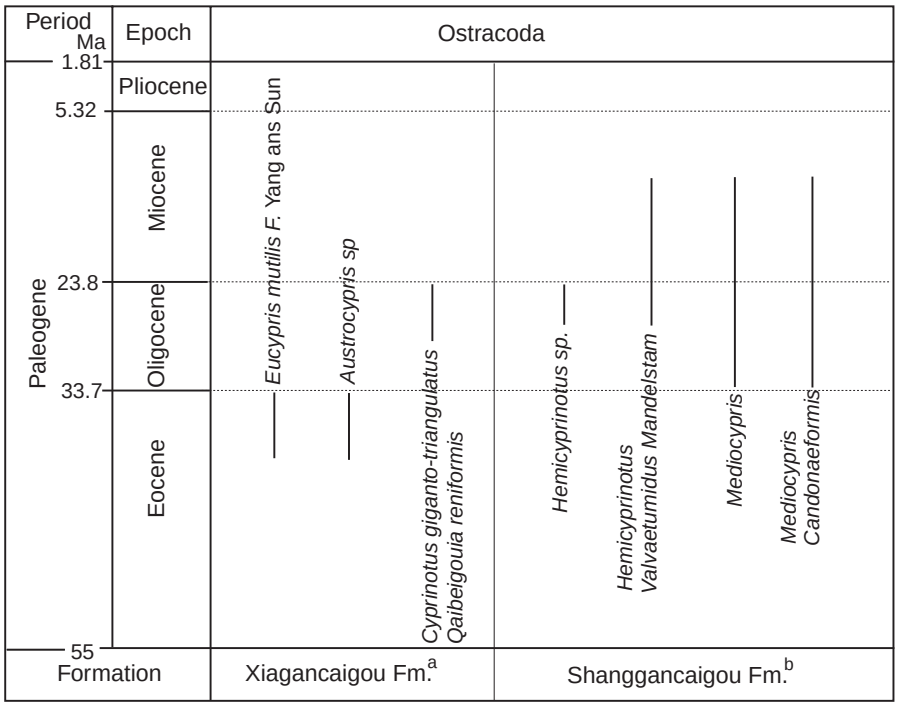


Fig. 3. The distribution of Ostracoda for the Shanggancaigou and Xiagancaigou Formations from the Qaidam Basin. a-from the studied section, b-from the Qaidam Basin.

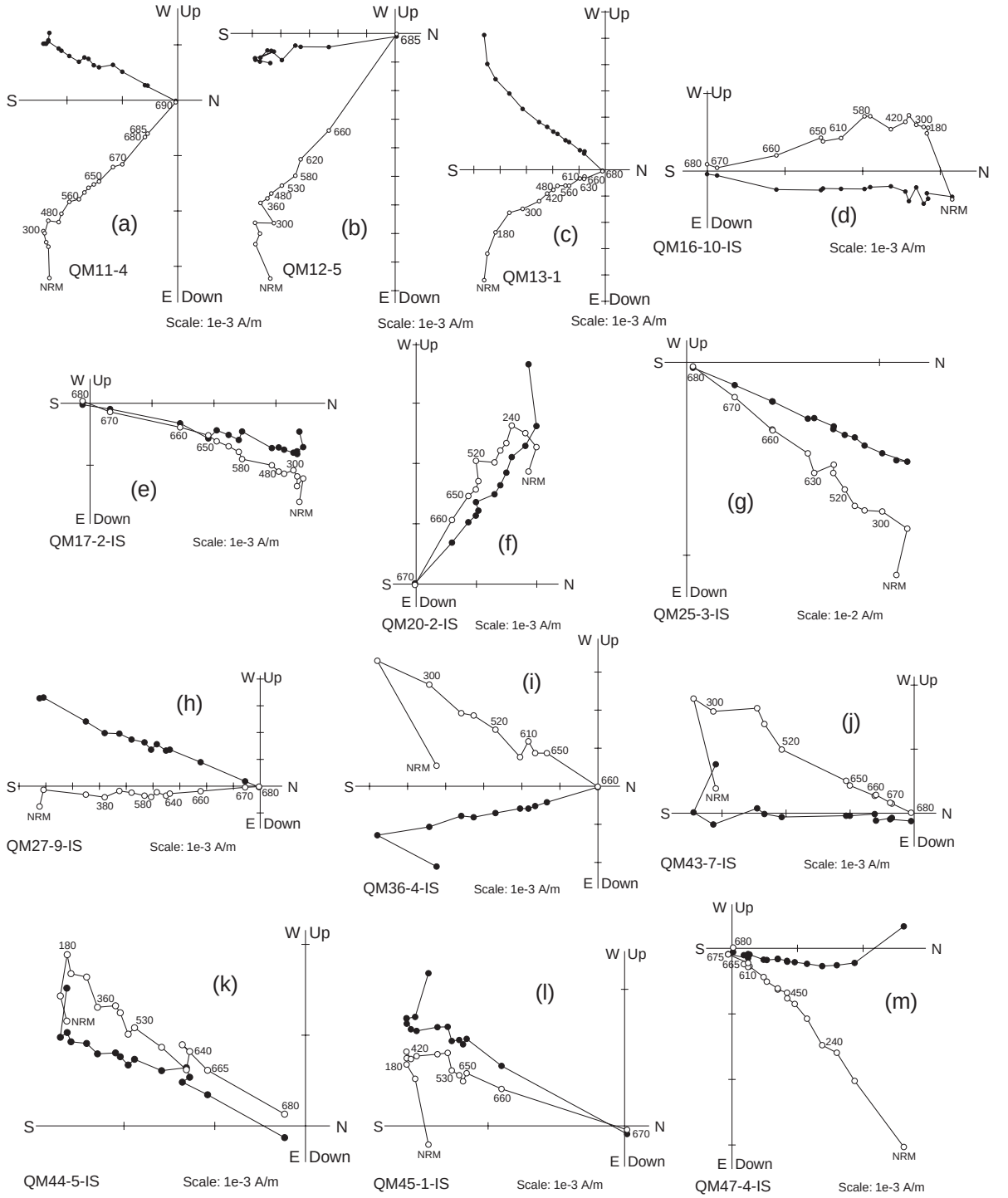


Fig. 4. Orthogonal vector plots of representative specimens from the studied section. Solid (open) symbols refer to the projection on the horizontal (vertical) plane in geographic coordinates. 200=200 °C. a–h: Xiagancaigou Formation., i–m: Shanggancaigou Formation.

The Shanggancaigou Formation is underlain unconformably by the Qigequan Formation that mainly consists of conglomerate up to 620-m thick. The presence of Ostracoda assemblage (①: *Limnocythere, Potamocypris, Eacypris, Limnocythere, Leucocythere, Cyclocypris*) at the lowest of the Qigequan Formation in the studied section indicates an age of Early Pleistocene (Fig. 7) [12].

The average sampling interval was about 2 m, but varied between 0.5 and 5 m, depending on the lithology. We have sampled 216 and 184 samples in the Xiagancaigou and Shanggancaigou Formations respectively. For the upper part of Shanggancaigou Formation, the conglomerates were impossible to sample, we therefore collected specimens from the intercalated siltstone or sandstone.

All paleomagnetic samples were drilled using a portable gasoline-powered core drill, and oriented with magnetic compasses. Sampling was mainly conducted at mudstone and sandstone. Samples were cut into 2.2 cm long cylinders for subsequent paleomagnetic analysis.

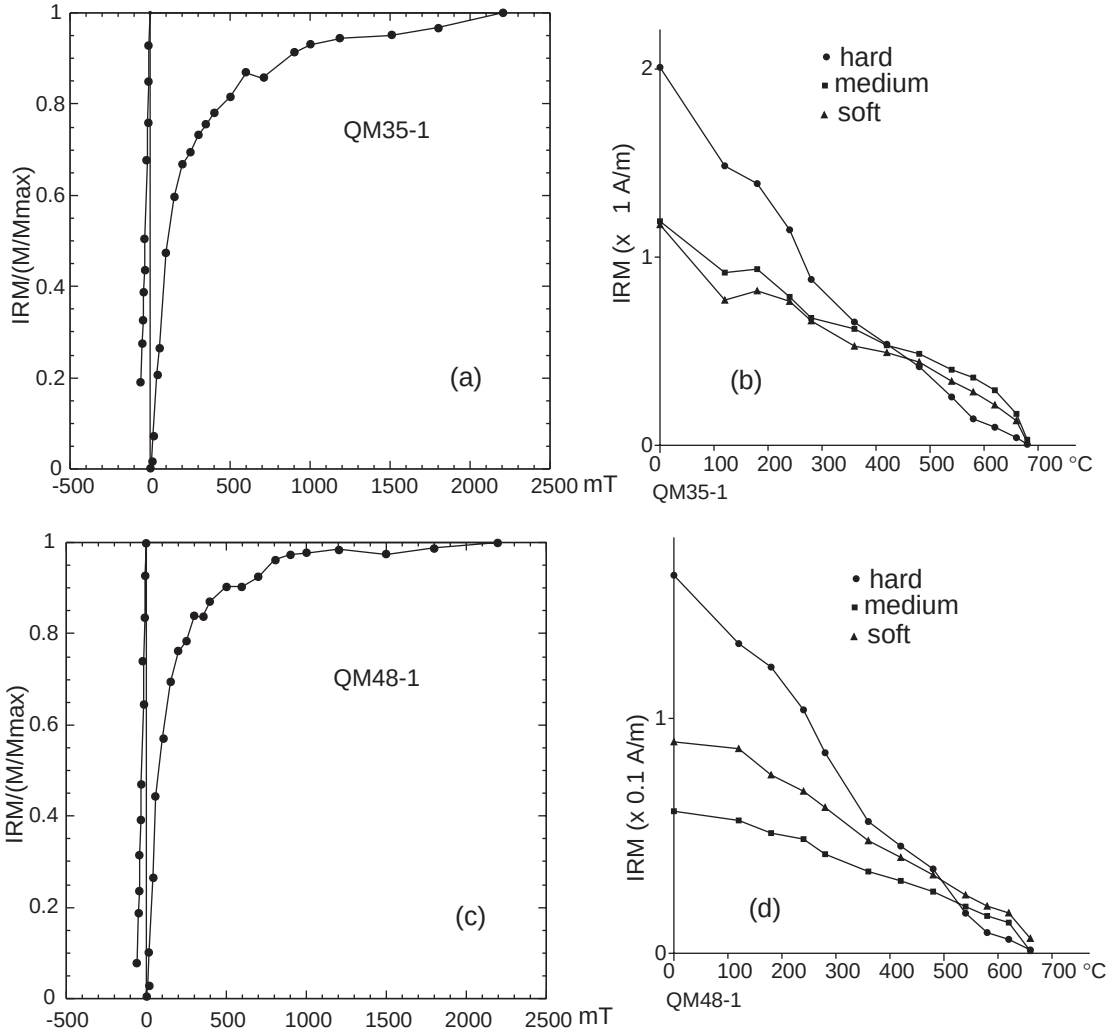


Fig. 5. Normalized isothermal remanent magnetization acquisition curves (a, c) and thermal demagnetization of a three-component IRM (b, d) of representative samples. The soft (<0.12 T), medium (0.12–0.4 T), and hard (0.4–1.2 T) components are shown during thermal demagnetization. Magnetic mineral is dominated by hematite (unblocking temperature at 680 °C).

3. Paleomagnetic analysis

All samples were subjected to stepwise thermal demagnetization in 17–21 steps in an ASC TD-48 oven with an internal residual field less than 10 nT. Remanent magnetization was measured with a 2 G cryogenic magnetometer at the Paleomagnetic Laboratory of the Institute of Geomechanics, CAGS in Beijing. The magnetometer is located inside a set of large Helmholtz coils that reduced the ambient geomagnetic field to around 300 nT. Magnetic components were determined by principal component analysis [15]. Site-mean paleomagnetic directions were calculated using Fisherian [16] statistics. In order to examine the magnetic carriers of the rocks, some representative samples were selected for the acquisition of isothermal remanent magnetization (IRM) and a back field demagnetization to estimate the coercivity.

Fig. 4 shows the Zijderveld plots of representative samples. About 4% of the total samples were discarded because their directions at higher temperature became erratic. Most samples possess two magnetic components: a low-temperature component (LTC) (generally below 300 °C) with random directions, another high-temperature component (HTC) (300–680 °C) that trends toward the origin. The high unblocking temperature of remanence (680 °C) implies the presence of hematite, which is also confirmed by IRM acquisitions and thermal demagnetization of three axes composite IRMs (Fig. 5). The HTC data are shown in Fig. 6. The average site-mean direction before tilt correction is D_g=2.9°, I_g=10.0°, κ_g=12.4, α_{95}=6.8°, and D_s=3.3°, I_s=43.9°, κ_s=36.6, α_{95}=3.8°, N=39 sites after tilt correction, which corresponds to a pole at 76.8° N, 258.8° E with dp/dm=3.0/4.8°. The site-mean directions of dual polarities not only pass a reversal test [17] with B classification (angle distance between site-mean directions of normal and reversed polarities is 5.1°, which is less than the critical angle of 7.9°), but also pass a fold test [18] ($\kappa s/\kappa g$=36.6/12.4=2.95>F(2*(n2−1), 2*(n1−1))=1.45) at the 95% confidence level. The Watson and Enkin [19] test gives an optimal concentration at 105% unfolding percent (with uncertainties ranging from 92.5% to 117.0%). Thus, we consider that the HTC magnetization most likely represents the primary remanence direction.

4. Correlation to the GPTS

The declination and inclination obtained from the ChRM of the specimens were used to calculate the latitude of virtual geomagnetic pole (VGP). The paleomagnetic declination and inclination, and VGP latitude from the studied section were used to illustrate the magnetic polarity of the stratigraphic level (Fig. 7). As discussed above, the Ostracoda assem-

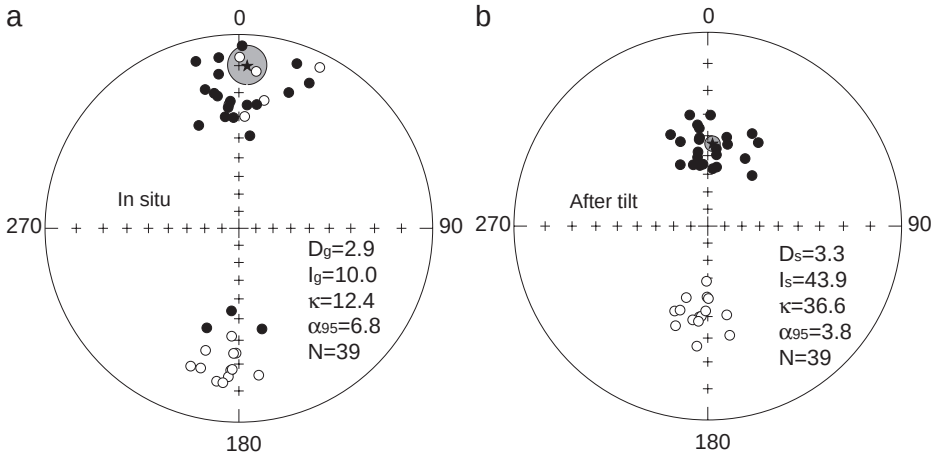


Fig. 6. Equal-area stereographic projections of site-mean directions of the high temperature components in (a) geographic and (b) stratigraphic coordinates, respectively. Lower (upper) hemisphere directions are marked with closed (open) symbols.

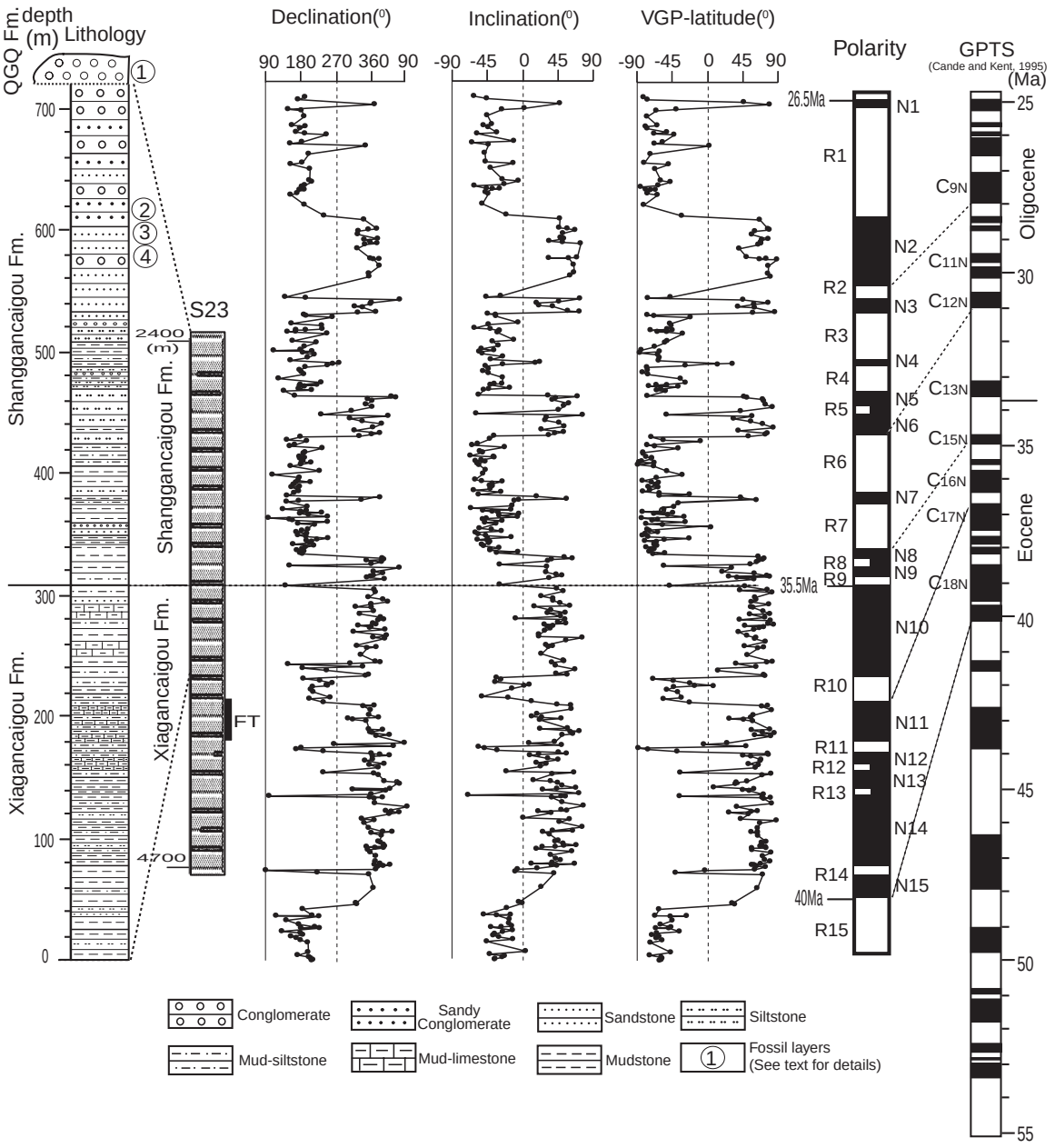


Fig. 7. Lithology and magnetostratigraphic results from the Hongsanhan section with declination and inclination and VGP latitude against stratigraphic level and correlation with the GPTS of Cande and Kent (1995). Stratigraphic occurrences of fossils found in the section are indicated by circled numbers. Magnetochrons defined by one sample are indicated by short-half bars.

blages give an age constraint of the Late Eocene to Oligocene for the Xiaganchaigou Formation [12–14], and the Oligocene to Middle Miocene for the Shanggancaigou Formation (Fig. 3). The overlying Qigequan Formation is Early Pleistocene in age [12]. In addition, a fission track (FT) dating of gypsum (53.68 ± 5.24 Ma) was reported in the lower part of the Xiagancaigou Formation from the drill core (No.

S23) [11] (Figs. 1 and 7). Thus, we suggest that the sampled Tertiary strata span roughly from 53.68 ± 5.24 Ma to Middle Miocene. Based on these age constraints and spacings of the observed polarity intervals, we can anchor the detailed magnetic polarity sequence to the documented Geomagnetic Polarity Time Scale (GPTS) of Cande and Kent [20] (Fig. 7).

Fifteen normal (N1–N15) and 15 reversed polarity zones (R1–R15) are clearly observed in the studied section. As shown in Fig. 7, the distinctive interval pattern of local magnetozones (N10–N15) in the Xiagancaigou Formation may correspond well to the subchrons (C16n to C18.2n) of the GPTS [20]. In the lower parts of the Shanggancaigou Formation, the distinctive short normal magnetozone (N7) interbedded with two relative long reversed polarities (R6 and R7) is easily correlated with the subchrons between C12r and C13r. Therefore, the upper magnetozones (N1–N6) in the upper parts of the Shanggancaigou Formation can be matched with most of the magnetic subchrons (C8.2n to C12n) of the GPTS. The magnetostratigraphic result suggests that the age of the Xiagancaigou Formation is >40 Ma to 35.5 Ma (Mid Eocene), and the age of the Shangancaigou Formation is between 35.5 Ma and 26.5 Ma (Late Eocene to Late Oligocene).

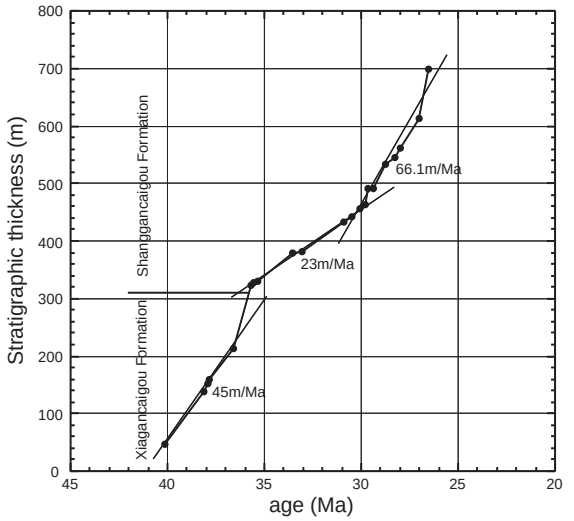


Fig. 8. Age versus stratigraphic thickness plot of the Hongsanhan section using the correlations of the magneozones shown in Fig. 7. Age constraints are determined by the polarity chrons.

5. Discussion

5.1. Accumulation rates and tectonic uplift of northern Tibetan plateau

Cenozoic uplift and erosion of the surrounding mountain ranges and the Tibetan plateau were responsible for the massive sediments deposited in the Qaidam Basin. Data on the stratigraphy, age and depositional style of the Mid-Eocene to Oligocene sediments in the Hongsanhan section may provide useful insights into the relationship between mountains uplift, rock denudation, and sedimentation. Fig. 7 illustrates the changes in lithofacies and the estimated age from this study. The sedimentation rate is estimated based on the thickness and age of sediments (Fig. 8). On the basis of upper and lower age bounds of the Mid-Eocene deposition, the Xiagancaigou Formation yielded a roughly uniform sedimentation rate (45 m/Ma). In contrast, the relationship of age and stratigraphic thickness for the Shanggancaigou Formation defines two distinctive deposition rates: 23 m/Ma from 35.5 to 30 Ma and 66.1 m/Ma from 30 to 26.5 Ma. This result suggests a major change in the lithofacies and accumulation rates occurred at around 30 Ma (near the bottom of N5). Depositional sediments are dominated by mudstone, siltstone, sandy mud-siltstone and mud-limestone, intercalated with gypsum layers prior to 30 Ma, which represent a lacustrine environment with a low paleodrainage gradient. The sedimentation rate was 45 m/Ma before 35.5 Ma, and decreased to 23 m/Ma during 35.5 to 30 Ma. However, the lithology is dominated by sandstone, intercalated with coarse conglomerate beds after 30 Ma. Most of these conglomerates are subrounded to rounded, but poorly sorted. In addition, the sedimentation rate increased abruptly to 66.1 m/Ma (Fig. 8).

Although the increase of conglomerate deposits resulted mainly from the erosion of mountains uplifting, local or regional climate change can also play an important role in regulating the conglomerate deposition in the studied area. More than 10 conglomerate layers were found in this studied section between 30 to 26.5 Ma. If climate played an important role in controlling the accumulation of conglomerate, there could have been at least 10 climatic cycles within

about 3.5 Ma. Studies on the pollen sequence and evaporitic deposits have suggested that a relative dry climate prevailed during the Eocene to Oligocene in the Qaidam Basin [21], which argued against the important role of climate in the deposition of such conglomerate layers. Instead, the abrupt changes in lithologies and abrupt increase in sedimentation rates in the studied section most likely reflect local thrusting and the early uplift of the northern Qaidam at about 30 Ma.

Previous studies also showed that the sedimentation rate increased after 30 Ma in the Qaidam Basin. For example, Métivier [22] suggested that the sedimentation rate increased around 30 Ma in the Qaidam Basin immediately to the north of the eastern Kunlun range. Several other studies [23,24] also documented an increase in sedimentation rate for the Qaidam Basin at the beginning of the Oligocene. These results and our data suggest the presence of at least some topographic relief along the northern edge of Tibetan plateau since about 30 Ma. The rapid increase in sedimentation rate in the Qaidam Basin could be a result of substantial uplift of the northern Tibetan plateau starting at about 30 Ma. In addition, field observations suggest that the northwest–southeast trending basement wedges along the northern Qaidam range were uplifted by thrusting tectonics around 30 Ma ago [25]. The uplift history of the northern Qaidam range is also supported by thermo-chronological results on the fault movement (East Kunlun fault) in the eastern Kunlun range, and from the northern and the southern edges of the Qaidam Basin [26,27]. Mock et al. (1999) [26] reported a cooling/unroofing event at around 30 Ma in the eastern Kunlun range by modeling K-feldspar $^{40}Ar/^{39}Ar$ release spectrum, and suggested that the eastern Kunlun underwent an important crustal thickening before 30 Ma. Recently, $^{40}Ar/^{39}Ar$ and fission track results from the northern and the southern edges of the Qaidam Basin also suggest that the northern Qaidam range was uplifted rapidly during the Mid-Oligocene to Miocene [27]. These results suggest that abrupt increase of sedimentation rate in the Qaidam Basin was resulted from rapid uplift events in the mountain ranges (northern Qaidam, east Kunlun) surrounding the Qaidam Basin at around 30 Ma, which may reflect the early growth of Tibetan plateau due to the penetration of India into Asia.

5.2. Limited rotation of the northwestern Qaidam Basin since the Eocene

The tilt-corrected mean direction is determined from the HTC components of 39 sites (Fig. 6). The mean observed declination is indistinguishable from the expected declination calculated from the Tertiary (30 Ma) Eurasian reference pole [28] ($-5.5 \pm 4.8^\circ$ at the reference point (38.6° N/91.9° E)) at the 95% confidence level, which implies insignificant rotation of the studied area since the Mid Eocene. In addition, no significant rotations have been observed with respect to Eurasia or Tarim in Tertiary sediments in the vicinity of the ATF including the northern Qaidam Basin [29,30], the interior Qaidam Basin [31], the southern Tarim Basin [29], and the Turfan Basin [32]. The lack of such rotations in the adjacent areas of the ATF indicates, on the one hand, that shear associated with the left-lateral slip may be localized on the fault zone that did not extend into the surrounding basins [33]. On the other hand, little curvature in the large-scale left-lateral slip of the ATF may be responsible for insignificant tectonic rotation of the Qaidam Block since the Mid Eocene.

The site-mean inclination shows a latitudinal difference ($15.5 \pm 3.9^\circ$) relative to the Tertiary Eurasian reference pole [28]. This result is similar to the anomalously shallow inclinations reported from Cretaceous and Tertiary rocks in the Qaidam and Tarim Basins by many authors [31,34–36]. The northward displacements relative to Siberia these low paleolatitudes are much larger than the estimates of intracontinental shortening within several mountain belts (Kunlun, Tian Shan, and Altay). Several hypotheses have been proposed to explain this inclination discrepancy, such as poor age control of sediments, synsedimentary and/or compaction-induced inclination shallowing, non-dipolar geomagnetic field geometry, tectonic shortening, or a poorly APWP constrained for Eurasia and non-rigidity of the Eurasian plate [34–40]. Some authors suggested that compaction unlikely affected the inclination of all red sandstones up to such large value (~20°) in Central Asia [40,41]. However, recent results of laboratory redepositional experiments suggest the shallow inclinations may result from depositional and/or compaction-caused inclination error in both red mudstone and sandstone [36,42,43]. More rock-magnetic and

paleomagnetic data from Cretaceous–Tertiary volcanic rocks are needed to resolve this problem of inclination shallowing.

6. Conclusion

The magnetostratigraphic study provides a detailed chronological control for the Eocene to Oligocene deposits in the northern Qaidam Basin. The age of the Xiagancaikou Formation is paleomagnetically dated as >40 to 35.5 Ma (Mid Eocene), and the age of the Shanggancaigou Formation is between 35.5–26.5 Ma (Late Eocene to Late-Mid Oligocene). Abrupt change in lithology as well as abrupt increase in sedimentation rate in the studied section were resulted mainly from the regional tectonic thrusting and induced uplift of the northern Qaidam at around 30 Ma as a far-field effect of the penetration of India into Asia. The paleomagnetic declination result indicates a slight counterclockwise rotation ($-5.5 \pm 4.8^{\circ}$) of the studied area relative to Eurasia since the Mid Eocene. Combined with previous paleomagnetic results obtained from the Tertiary sediments in the adjacent areas of the ATF, these results suggest that little curvature of the large-scale left-lateral slip of the AFT may be responsible for insignificant tectonic rotation of the Qaidam Block since the Mid Eocene.

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